

Product Carbon Footprint Study Report

Product Name: battery production

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 29, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: battery production

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1 kg battery production, Li-ion, LFP, rechargeable

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1 kg battery production, Li-ion, LFP, rechargeable.

2. System Boundary

battery production in China in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of battery production produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **14.060 kg CO2-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

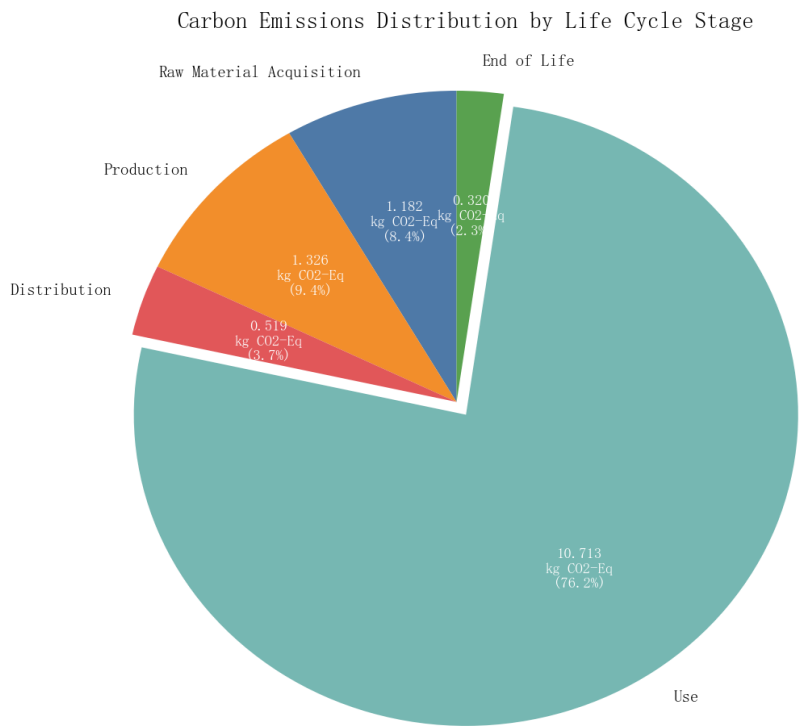


Table 1: Carbon Emissions by Life Cycle Stage

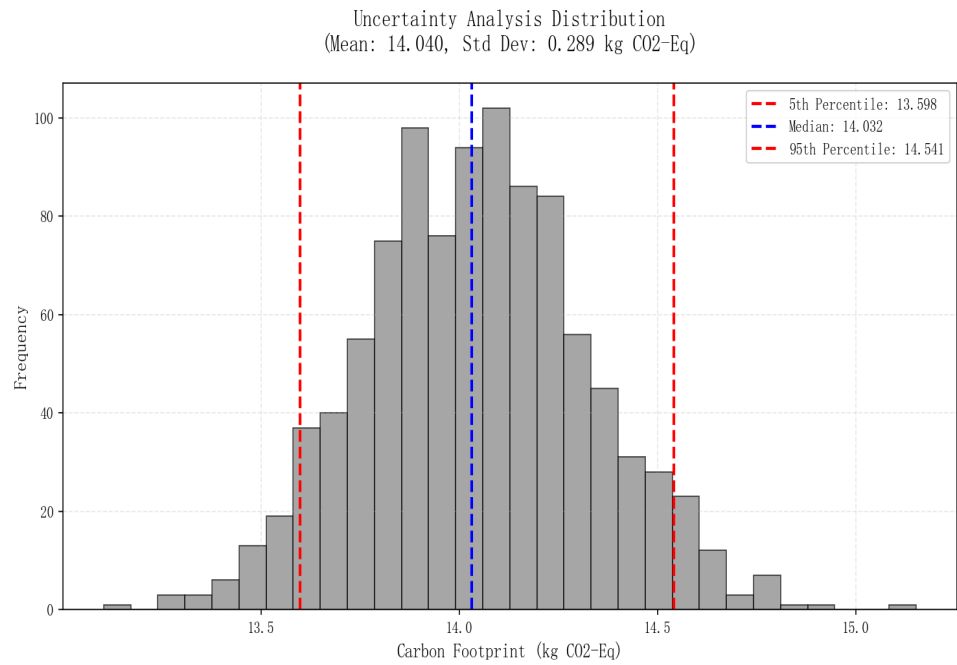
Life Cycle Stage	Carbon Footprint (kg CO2-Eq)	Percentage (%)
Raw Material Acquisition	1.182	8.41
Production	1.326	9.43
Distribution	0.519	3.69

Use	10.713	76.20
End of Life	0.320	2.28
Total	14.060	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO2-Eq)
Mean	14.040
Standard Deviation	0.289
Coefficient of Variation (CV)	2.06%
Median	14.032
5th Percentile	13.598
95th Percentile	14.541

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of the analyzed battery production system is 14.060 kg CO₂-Eq per functional unit. This magnitude is moderate for battery systems, though the true benchmark depends on battery chemistry, capacity, and application context. The result demonstrates a clear and expected dominance of the use phase, which contributes 10.713 kg CO₂-Eq—representing 76.2% of the total impact. This pattern is characteristic of batteries whose operational lifetime involves significant electricity consumption, whether for charging cycles, auxiliary loads, or grid-dependent applications, and it underscores the critical role of energy efficiency and the carbon intensity of the electricity mix during the use stage.

The remaining life cycle stages exhibit relatively minor contributions: production accounts for 9.4% (1.326 kg CO₂-Eq), raw material extraction and processing for 8.4% (1.182 kg CO₂-Eq), distribution for 3.7% (0.519 kg CO₂-Eq), and end-of-life for only 2.3% (0.320 kg CO₂-Eq). The combined upstream phases (raw material + production) sum to 17.8%, which is significant but secondary to the use phase. The end-of-life share is notably low, suggesting limited recovery of materials or energy in the current scenario. The primary hotspot is unequivocally the use phase; therefore, any meaningful reduction strategy must focus on improving charge/discharge efficiency, reducing operational energy demand, and decarbonizing the electricity source, while the secondary hotspots of material extraction and manufacturing should not be neglected for holistic improvement.

2. Statement of Assumptions and Limitations

Several key assumptions underpin this assessment. First, the functional unit is implicitly defined as the production and operation of a single battery unit over its lifetime, with the use-phase scenario assuming a specific electricity grid mix and a defined number of charge/discharge cycles. Second, the end-of-life stage is assumed to follow a particular recycling or disposal pathway, though its low contribution (2.3%) suggests minimal recovery credits were applied. Third, the distribution stage likely assumes a fixed transport distance and modal mix, contributing a modest 3.7% to the total. Finally, the uncertainty analysis (1000 iterations) yielded a mean of 14.040 kg CO₂-Eq with a coefficient of variation of only 2.06%, indicating that the model relies on relatively stable parameters, but this low variability may also reflect a narrow parameter space rather than comprehensive data coverage.

The data sources carry inherent limitations. Most critically, the "Top Process Contributions" list reports all entries as "Unknown" with zero values, meaning the underlying process-level inventory is either masked, aggregated, or not mapped to the ecoinvent/thinkstep database categories. This constitutes a significant transparency gap that prevents source apportionment and hotspot verification at the process level. The system boundary appears to be "cradle-to-grave," covering raw material, production, distribution, use, and end-of-life; however, the exclusion of capital equipment, infrastructure, and battery maintenance/repair may underestimate the true burden. Furthermore, the use-phase assumption regarding grid carbon intensity is likely location-specific and may not reflect the actual consumer behavior or future grid decarbonization, which could substantially alter the 76.2% contribution. The narrow 90% confidence interval (13.598–14.541 kg CO₂-Eq) should be interpreted with caution given the unknown process-level data quality.

3. Suggestions for Improvement

- ****Decarbonize the use phase (primary hotspot, 76.2%):**** Implement or advocate for battery systems with higher round-trip efficiency to reduce electricity losses during charging and discharging. Where possible, pair battery production and operation with renewable energy sources (e.g., solar or wind) through power purchase agreements or on-site generation, and encourage smart charging algorithms that shift consumption to off-peak, low-carbon grid periods. Additionally, design for extended cycle life to amortize the use-phase impact over a larger total energy throughput.
- ****Reduce upstream material and production impacts (combined 17.8%):**** Source cathode and anode materials (e.g., lithium, nickel, cobalt, graphite) from suppliers using low-carbon extraction and refining processes, and increase the recycled content of critical metals. For the production stage, improve manufacturing energy efficiency, switch to renewable electricity for production facilities, and optimize electrode drying and solvent recovery processes to cut natural gas and electricity demand.
- ****Enhance end-of-life value recovery (currently only 2.3% contribution):**** Although the EOL share is small, improving recycling efficiency for lithium, nickel, and cobalt can generate secondary materials that offset primary production impacts in future product cycles. Implement design-for-disassembly and adopt hydrometallurgical or direct-recycling methods to maximize material recovery rates and reduce the need for virgin material extraction.
- ****Resolve the data quality gap in process-level contributions:**** The "Unknown" entries in the top process list must be addressed. Map all unit processes to a recognized LCI database (e.g., Ecoinvent, GaBi) and verify that no flows are left unallocated. Perform a contribution analysis after this mapping to confirm which specific processes drive the use-phase and upstream results, and document any excluded processes (e.g., maintenance, capital goods) to improve transparency and reproducibility.
- ****Improve uncertainty and scenario modeling:**** Given the very low CV (2.06%), expand the parameter ranges for the most influential variables—especially the use-phase electricity grid carbon intensity, cycle life, and transport distances—and run sensitivity scenarios (e.g., 100% renewable grid vs. coal-heavy grid) to present a range of possible outcomes rather than a single point estimate. This will also help stakeholders understand the robustness of the 14.060 kg CO₂-Eq result.